# Model types: ODE, DDE, stochastic, and discrete

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## Introduction

A single Petri net can generate odin2 code for four different model types, following the framework of AlgebraicDynamics.jl. Each type uses the same stoichiometry but differs in how the dynamics are computed.

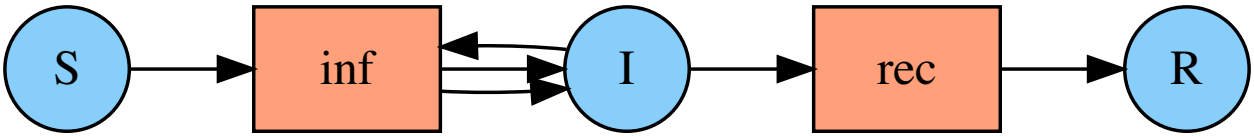
| Type | odin2 keyword | Time | Randomness |
|---|---|---|---|
| **ODE** | `deriv()` | Continuous | Deterministic |
| **DDE** | `deriv()` + `delay()` | Continuous | Deterministic |
| **Stochastic** | `update()` + `Binomial()` | Discrete | Stochastic |
| **Discrete** | `update()` | Discrete | Deterministic |

```r
library(algebraicodin)
```

We use the standard SIR model throughout:

```r
sir <- labelled_petri_net(
  c("S", "I", "R"),
  inf = c("S", "I") %=>% c("I", "I"),
  rec = "I" %=>% "R"
)
```

```r
plot_petri(sir)
```




## ODE (continuous deterministic)

The default model type. Uses `deriv()` to define continuous derivatives solved by numerical integration:

```r
cat(pn_to_odin(sir, "ode"))
#> ## Auto-generated by algebraicodin
#>
#> ## Parameters
#> inf <- parameter()
#> rec <- parameter()
#>
#> ## Initial conditions
#> S0 <- parameter()
#> I0 <- parameter()
#> R0 <- parameter()
```

```
#>
#> initial(S) <- S0
#> initial(I) <- I0
#> initial(R) <- R0
#>
#> ## Transition rates
#> rate_inf <- inf * S * I
#> rate_rec <- rec * I
#>
#> ## Derivatives
#> deriv(S) <- -rate_inf
#> deriv(I) <- rate_inf - rate_rec
#> deriv(R) <- rate_rec
```

```
gen <- odin2::odin(pn_to_odin(sir, "ode"))
#> i Using cached generator
pars <- list(inf = 0.0005, rec = 0.25, S0 = 990, I0 = 10, R0 = 0)
sys <- dust2::dust_system_create(gen, pars, n_particles = 1)
dust2::dust_system_set_state_initial(sys)
t_ode <- seq(0, 40, by = 0.1)
y_ode <- dust2::dust_system_simulate(sys, t_ode)
```

### Hand-written comparison

```
ode_handwritten <- "
inf <- parameter()
rec <- parameter()
S0 <- parameter()
I0 <- parameter()
R0 <- parameter()
initial(S) <- S0
initial(I) <- I0
initial(R) <- R0
deriv(S) <- -inf * S * I
deriv(I) <- inf * S * I - rec * I
deriv(R) <- rec * I
"
gen_hw <- odin2::odin(ode_handwritten)
#> v Wrote 'DESCRIPTION'
#> v Wrote 'NAMESPACE'
#> v Wrote 'R/dust.R'
#> v Wrote 'src/dust.cpp'
#> v Wrote 'src/Makevars'
#> i 13 functions decorated with [[cpp11::register]]
#> v generated file 'cpp11.R'
#> v generated file 'cpp11.cpp'
#> i Re-compiling odin.system4ba2abf0
#> -- R CMD INSTALL ----------------------------------------------------------------------------------------
#>   -  installing *source* package ‘odin.system4ba2abf0’ ...
#>    ** this is package ‘odin.system4ba2abf0’ version ‘0.0.1’
#>      ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
```

```
#>      using SDK: 'MacOSX15.5.sdk'
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:77:
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/contin
#>       34 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                          ^
#>    dust.cpp:81:20: note: in instantiation of function template specialization 'dust2::r::dust2_conti
#>       81 |   return dust2::r::dust2_continuous_alloc<odin_system>(r_pars, r_time, r_time_control, r_
#>          |                    ^
#>      2 warnings generated.
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mn1n7x556c2v4w80000gn/T/RtmpMjuTY0/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
#>   -  DONE (odin.system4ba2abf0)
#>
#> i Loading odin.system4ba2abf0
sys_hw <- dust2::dust_system_create(gen_hw, pars, n_particles = 1)
dust2::dust_system_set_state_initial(sys_hw)
y_hw <- dust2::dust_system_simulate(sys_hw, t_ode)
cat("ODE max diff:", max(abs(y_ode - y_hw)), "\n")
#> ODE max diff: 1.136868e-13
```

## DDE (delay differential equations)

DDEs introduce delays into the dynamics. In epidemiology, this can model a fixed incubation period or delayed immune response.

The `delays` argument specifies which transitions use delayed values and the delay time (`tau`):

```
dde_code <- pn_to_odin(sir, "dde", delays = list(
  inf = list(tau = 5, species = "I")
))
cat(dde_code)
#> ## Auto-generated by algebraicodin
#>
#> ## Parameters
#> inf <- parameter()
#> rec <- parameter()
#>
#> ## Initial conditions
#> S0 <- parameter()
#> I0 <- parameter()
#> R0 <- parameter()
#>
#> initial(S) <- S0
#> initial(I) <- I0
```

```
#> initial(R) <- R0
#>
#> ## Delayed variables
#> I_delayed_inf <- delay(I, 5)
#>
#> ## Transition rates
#> rate_inf <- inf * S * I_delayed_inf
#> rate_rec <- rec * I
#>
#> ## Derivatives
#> deriv(S) <- -rate_inf
#> deriv(I) <- rate_inf - rate_rec
#> deriv(R) <- rate_rec
```

The infection rate now uses `delay(I, 5)` instead of the current value of I, meaning that infectiousness depends on the state 5 time units in the past.

### Hand-written comparison

```
dde_handwritten <- "
inf <- parameter()
rec <- parameter()
S0 <- parameter()
I0 <- parameter()
R0 <- parameter()
initial(S) <- S0
initial(I) <- I0
initial(R) <- R0
I_delayed <- delay(I, 5)
deriv(S) <- -inf * S * I_delayed
deriv(I) <- inf * S * I_delayed - rec * I
deriv(R) <- rec * I
"
```

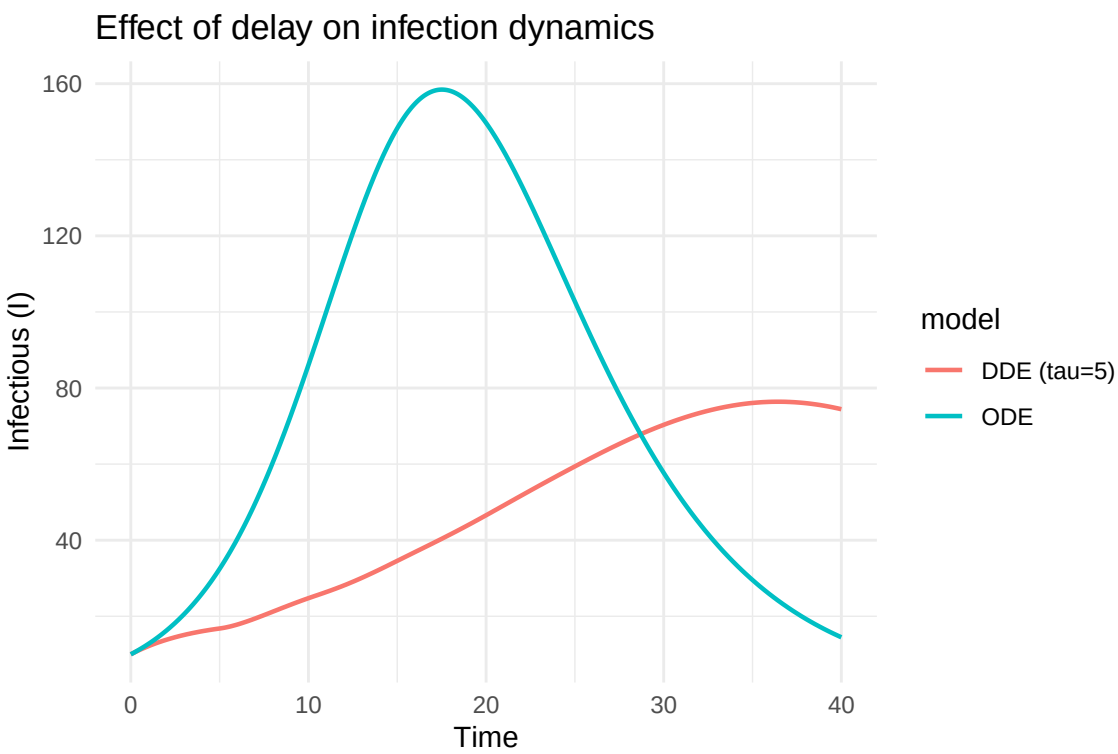
```
gen_dde <- odin2::odin(dde_code)
#> v Wrote 'DESCRIPTION'
#> v Wrote 'NAMESPACE'
#> v Wrote 'R/dust.R'
#> v Wrote 'src/dust.cpp'
#> v Wrote 'src/Makevars'
#> i 13 functions decorated with [[cpp11::register]]
#> v generated file 'cpp11.R'
#> v generated file 'cpp11.cpp'
#> i Re-compiling odin.systemdb69a213
#> -- R CMD INSTALL ------------------------------------------------------------------------------
#>   -  installing *source* package ‘odin.systemdb69a213’ ...
#>    ** this is package ‘odin.systemdb69a213’ version ‘0.0.1’
#>      ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
#>      using SDK: ‘MacOSX15.5.sdk’
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:84:
```

```
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/contin
#>       34 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                     ^
#>    dust.cpp:88:20: note: in instantiation of function template specialization 'dust2::r::dust2_conti
#>     88 |   return dust2::r::dust2_continuous_alloc<odin_system>(r_pars, r_time, r_time_control, r_n_
#>          |                    ^
#>      2 warnings generated.
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mn1n7x556c2v4w80000gn/T/RtmpMjuTY0/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
#>   -  DONE (odin.systemdb69a213) (463ms)
#>
#> i Loading odin.systemdb69a213
pars_dde <- list(inf = 0.0005, rec = 0.25, S0 = 990, I0 = 10, R0 = 0)
sys_dde <- dust2::dust_system_create(gen_dde, pars_dde, n_particles = 1)
dust2::dust_system_set_state_initial(sys_dde)
y_dde <- dust2::dust_system_simulate(sys_dde, t_ode)

gen_hw2 <- odin2::odin(dde_handwritten)
#> v Wrote 'DESCRIPTION'
#> v Wrote 'NAMESPACE'
#> v Wrote 'R/dust.R'
#> v Wrote 'src/dust.cpp'
#> v Wrote 'src/Makevars'
#> i 13 functions decorated with [[cpp11::register]]
#> v generated file 'cpp11.R'
#> v generated file 'cpp11.cpp'
#> i Re-compiling odin.system87a9fd7a
#> -- R CMD INSTALL ----------------------------------------------------------------------------------
#>   -  installing *source* package ‘odin.system87a9fd7a’ ...
#>    ** this is package ‘odin.system87a9fd7a’ version ‘0.0.1’
#>      ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
#>      using SDK: ‘MacOSX15.5.sdk’
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:82:
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>     60 |       std::ceil(std::abs(::unif_rand        60 |       std::ceil(std::abs(::unif_rand()) * 
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/contin
```

```
#>       34 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                          ^
#>    dust.cpp:86:20: note: in instantiation of function template specialization 'dust2::r::dust2_conti
#>     86 |   return dust2::r::dust2_continuous_alloc<odin_system>(r_pars, r_time, r_time_control, r_n_
#>          |                      ^
#>      2 warnings generated.
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mn1n7x556c2v4w80000gn/T/RtmpMjuTYO/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
#>   -  DONE (odin.system87a9fd7a) (528ms)
#>
#> i Loading odin.system87a9fd7a
sys_hw2 <- dust2::dust_system_create(gen_hw2, pars_dde, n_particles = 1)
dust2::dust_system_set_state_initial(sys_hw2)
y_hw2 <- dust2::dust_system_simulate(sys_hw2, t_ode)
cat("DDE max diff:", max(abs(y_dde - y_hw2)), "\n")
#> DDE max diff: 1.136868e-13
```

```
library(ggplot2)
df <- data.frame(
  time = rep(t_ode, 2),
  I = c(y_ode[2, ], y_dde[2, ]),
  model = rep(c("ODE", "DDE (tau=5)"), each = length(t_ode))
)
ggplot(df, aes(time, I, colour = model)) +
  geom_line(linewidth = 0.8) +
  labs(title = "Effect of delay on infection dynamics",
       x = "Time", y = "Infectious (I)") +
  theme_minimal()
```



The delay shifts the epidemic peak later and reduces its height, as the infection rate responds to historical rather than current infectiousness.

## Stochastic (discrete-time)

The stochastic model uses `Binomial()` draws for each transition, modelling demographic stochasticity:

```
cat(pn_to_odin(sir, "stochastic"))
#> ## Auto-generated by algebraicodin
#>
#> ## Parameters
#> inf <- parameter()
#> rec <- parameter()
#>
#> ## Initial conditions
#> S0 <- parameter()
#> I0 <- parameter()
#> R0 <- parameter()
#>
#> initial(S) <- S0
#> initial(I) <- I0
#> initial(R) <- R0
#>
#> ## Transition probabilities and draws
#> p_inf <- 1 - exp(-(inf * I) * dt)
#> n_inf <- Binomial(S, p_inf)
#> p_rec <- 1 - exp(-(rec) * dt)
#> n_rec <- Binomial(I, p_rec)
#>
#> ## Update equations
#> update(S) <- S - n_inf
#> update(I) <- I + n_inf - n_rec
#> update(R) <- R + n_rec
```

Key features: - Transition probabilities: $p = 1 - e^{-\text{rate} \cdot dt}$ - Draws: `Binomial(substrate, probability)` - Update equations use draws instead of rates

### Hand-written comparison

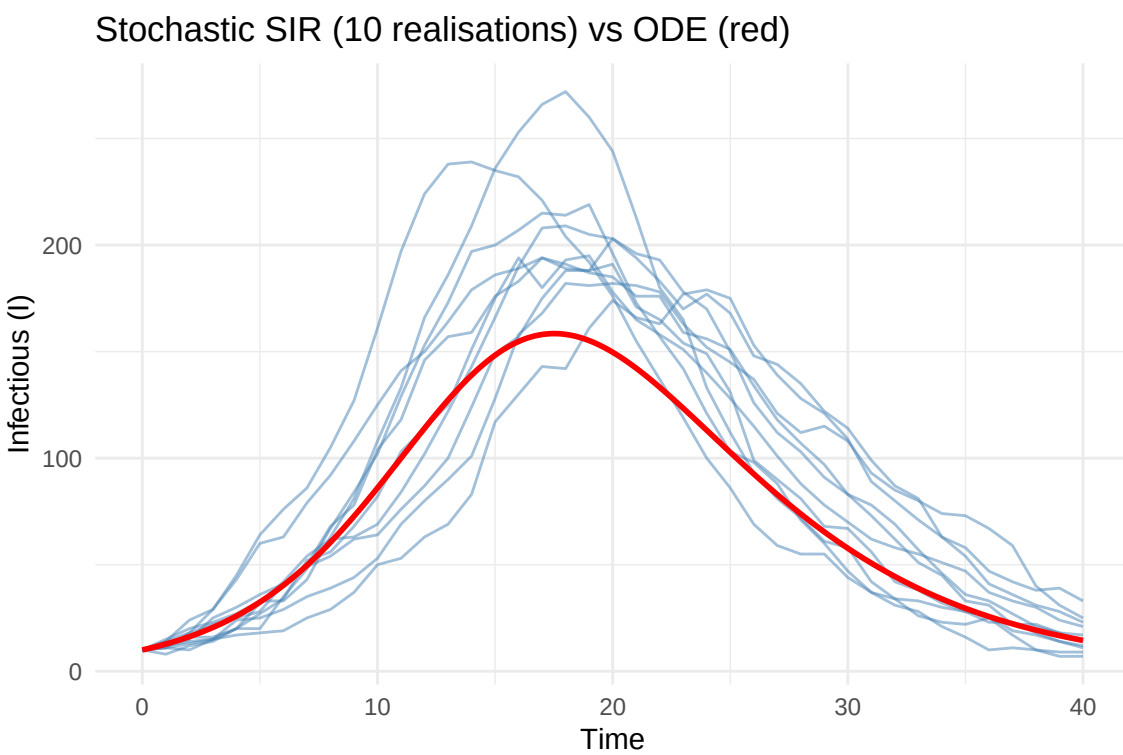
```
stoch_handwritten <- "
inf <- parameter()
rec <- parameter()
S0 <- parameter()
I0 <- parameter()
R0 <- parameter()
initial(S) <- S0
initial(I) <- I0
initial(R) <- R0
p_inf <- 1 - exp(-(inf * I) * dt)
n_inf <- Binomial(S, p_inf)
p_rec <- 1 - exp(-(rec) * dt)
n_rec <- Binomial(I, p_rec)
update(S) <- S - n_inf
update(I) <- I + n_inf - n_rec
```

```
update(R) <- R + n_rec
"
```

```
gen_stoch <- odin2::odin(pn_to_odin(sir, "stochastic"))
#> v Wrote 'DESCRIPTION'
#> v Wrote 'NAMESPACE'
#> v Wrote 'R/dust.R'
#> v Wrote 'src/dust.cpp'
#> v Wrote 'src/Makevars'
#> i 12 functions decorated with [[cpp11::register]]
#> v generated file 'cpp11.R'
#> v generated file 'cpp11.cpp'
#> i Re-compiling odin.system1c0df591
#> -- R CMD INSTALL ------------------------------------------------------------------------------------
#>   -  installing *source* package ‘odin.system1c0df591’ ...
#>      ** this is package ‘odin.system1c0df591’ version ‘0.0.1’
#>    ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
#>      using SDK: ‘MacOSX15.5.sdk’
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:82:
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/discre
#>       41 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                 ^
#>    dust.cpp:86:20: note: in instantiation of function template specialization 'dust2::r::dust2_discr
#>       86 |   return dust2::r::dust2_discrete_alloc<odin_system>(r_pars, r_time, r_time_control, r_n_
#>          |                    ^
#>      2 warnings generated.
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mn1n7x556c2v4w80000gn/T/RtmpMjuTY0/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
#>   -  DONE (odin.system1c0df591) (402ms)
#>
#> i Loading odin.system1c0df591
pars_stoch <- list(inf = 0.0005, rec = 0.25,
                   S0 = 990, I0 = 10, R0 = 0)
sys_stoch <- dust2::dust_system_create(gen_stoch, pars_stoch, n_particles = 10,
                                       seed = 42, dt = 1)
dust2::dust_system_set_state_initial(sys_stoch)
t_disc <- seq(0L, 40L, by = 1L)
y_stoch <- dust2::dust_system_simulate(sys_stoch, t_disc)

# Plot multiple realisations
dfs <- do.call(rbind, lapply(1:10, function(p) {
  data.frame(time = t_disc, I = y_stoch[2, p, ], particle = p)
```

```r
}))
ggplot(dfs, aes(time, I, group = particle)) +
  geom_line(alpha = 0.5, colour = "steelblue") +
  geom_line(data = data.frame(time = t_ode, I = y_ode[2, ]),
            aes(group = NULL), colour = "red", linewidth = 1) +
  labs(title = "Stochastic SIR (10 realisations) vs ODE (red)",
       x = "Time", y = "Infectious (I)") +
  theme_minimal()
```




### Discrete deterministic (Euler method)

The discrete deterministic model uses a forward Euler step, useful for comparison with the stochastic model or when discrete-time dynamics are desired:

```r
cat(pn_to_odin(sir, "discrete"))
#> ## Auto-generated by algebraicodin
#>
#> ## Parameters
#> inf <- parameter()
#> rec <- parameter()
#>
#> ## Initial conditions
#> S0 <- parameter()
#> I0 <- parameter()
#> R0 <- parameter()
#>
#> initial(S) <- S0
#> initial(I) <- I0
#> initial(R) <- R0
#>
```

```
#> ## Transition rates (deterministic Euler step)
#> rate_inf <- inf * S * I
#> rate_rec <- rec * I
#>
#> ## Update equations
#> update(S) <- S - dt * rate_inf
#> update(I) <- I + dt * rate_inf - dt * rate_rec
#> update(R) <- R + dt * rate_rec
```

**Hand-written comparison**

```
disc_handwritten <- "
inf <- parameter()
rec <- parameter()
S0 <- parameter()
I0 <- parameter()
R0 <- parameter()
initial(S) <- S0
initial(I) <- I0
initial(R) <- R0
rate_inf <- inf * S * I
rate_rec <- rec * I
update(S) <- S - dt * rate_inf
update(I) <- I + dt * rate_inf - dt * rate_rec
update(R) <- R + dt * rate_rec
"
gen_disc <- odin2::odin(pn_to_odin(sir, "discrete"))
#> v Wrote 'DESCRIPTION'
#> v Wrote 'NAMESPACE'
#> v Wrote 'R/dust.R'
#> v Wrote 'src/dust.cpp'
#> v Wrote 'src/Makevars'
#> i 12 functions decorated with [[cpp11::register]]
#> v generated file 'cpp11.R'
#> v generated file 'cpp11.cpp'
#> i Re-compiling odin.system7dcfd95e
#> -- R CMD INSTALL ------------------------------------------------------------------------------------
#>   -  installing *source* package ‘odin.system7dcfd95e’ ...
#>    ** this is package ‘odin.system7dcfd95e’ version ‘0.0.1’
#>      ** using staged installation
#>      ** libs
#>      using C++ compiler: ‘Homebrew clang version 21.1.5’
#>      using SDK: ‘MacOSX15.5.sdk’
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      clang++ -arch arm64 -std=gnu++17 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG
#>      In file included from dust.cpp:80:
#>    In file included from /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/random
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
#>      /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/monty/include/monty/r/rand
#>       60 |       std::ceil(std::abs(::unif_rand()) * std::numeric_limits<size_t>::max());
#>          |                                          ~ ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
```

```
#>    /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/library/dust2/include/dust2/r/discre
#>       41 |   auto seed = monty::random::r::as_rng_seed<rng_state_type>(r_seed);
#>          |                                          ^
#>    dust.cpp:84:20: note: in instantiation of function template specialization 'dust2::r::dust2_discr
#>     84 |         84 |   return dust2::r::dust2_discrete_alloc<odin_system>(r_pars, r_time, r_time_co
#>          |                     ^
#>      2 warnings generated.
#>      clang++ -arch arm64 -std=gnu++17 -dynamiclib -Wl,-headerpad_max_install_names -undefined dynami
#>      installing to /private/var/folders/yh/30rj513j6mn1n7x556c2v4w80000gn/T/RtmpMjuTYO/devtools_inst
#>      ** checking absolute paths in shared objects and dynamic libraries
#>   -  DONE (odin.system7dcfd95e) (433ms)
#>
#> i Loading odin.system7dcfd95e
gen_hw3 <- odin2::odin(disc_handwritten)
#> i Using cached generator

pars_disc <- list(inf = 0.0005, rec = 0.25,
                  S0 = 990, I0 = 10, R0 = 0)
sys_disc <- dust2::dust_system_create(gen_disc, pars_disc, n_particles = 1, dt = 1)
dust2::dust_system_set_state_initial(sys_disc)
t_int <- seq(0L, 40L, by = 1L)
y_disc <- dust2::dust_system_simulate(sys_disc, t_int)

sys_hw3 <- dust2::dust_system_create(gen_hw3, pars_disc, n_particles = 1, dt = 1)
dust2::dust_system_set_state_initial(sys_hw3)
y_hw3 <- dust2::dust_system_simulate(sys_hw3, t_int)
cat("Discrete max diff:", max(abs(y_disc - y_hw3)), "\n")
#> Discrete max diff: 0
```

## ResourceSharer and Machine types

For programmatic manipulation of dynamical systems (rather than code generation), algebraicodin provides `ResourceSharer` and `Machine` classes following AlgebraicDynamics.jl:

```
rs <- petri_to_continuous(sir)
rs@system_type
#> [1] "continuous"
rs@nstates
#> [1] 3
rs@state_names
#> [1] "S" "I" "R"
```

Evaluate the vector field at a given state:

```
u <- c(990, 10, 0)
p <- list(inf = 0.0005, rec = 0.25)
eval_dynamics(rs, u, p, t = 0)
#> [1] -4.95  2.45  2.50
```

## Euler approximation

Convert a continuous system to discrete via forward Euler:

```
rs_disc <- euler_approx(rs, h = 0.1)
rs_disc@system_type
```

```r
#> [1] "discrete"

# One Euler step
u_next <- eval_dynamics(rs_disc, u, p, t = 0)
cat("u_next:", u_next, "\n")
#> u_next: 989.505 10.245 0.25
cat("u + 0.1 * f(u):", u + 0.1 * eval_dynamics(rs, u, p, t = 0), "\n")
#> u + 0.1 * f(u): 989.505 10.245 0.25
```

### All four types from one Petri net

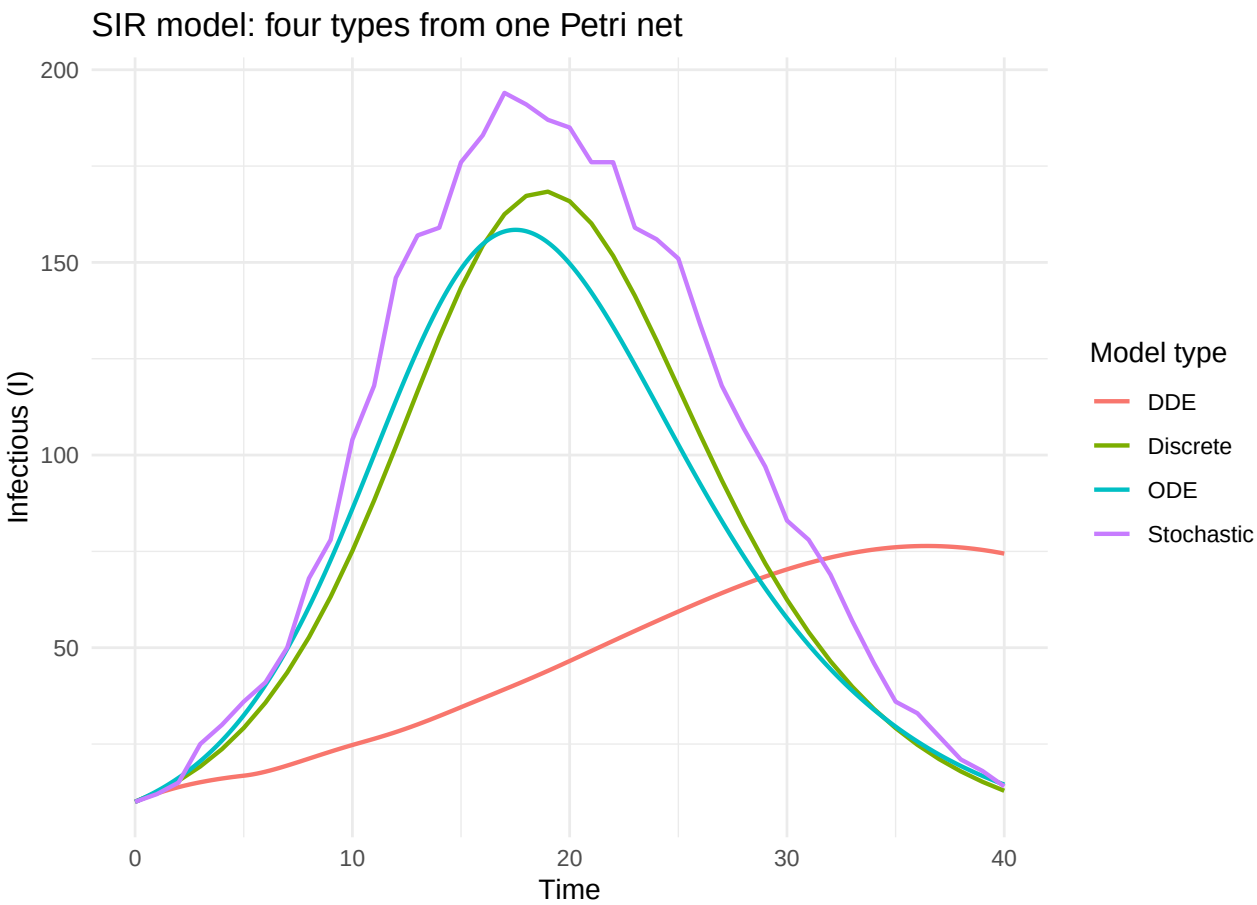
```r
# Run all four model types
run_model <- function(code, pars, times, type) {
  gen <- odin2::odin(code)
  is_discrete <- type %in% c("Stochastic", "Discrete")
  args <- list(generator = gen, pars = pars, n_particles = 1L,
               deterministic = (type != "Stochastic"), seed = 42L)
  if (is_discrete) args$dt <- 1
  sys <- do.call(dust2::dust_system_create, args)
  dust2::dust_system_set_state_initial(sys)
  y <- dust2::dust_system_simulate(sys, times)
  data.frame(time = times, I = y[2, ], type = type)
}

pars_cont <- list(inf = 0.0005, rec = 0.25, S0 = 990, I0 = 10, R0 = 0)
pars_disc <- pars_cont

dde_delays <- list(inf = list(tau = 5, species = "I"))

df_all <- rbind(
  run_model(pn_to_odin(sir, "ode"), pars_cont, t_ode, "ODE"),
  run_model(pn_to_odin(sir, "dde", delays = dde_delays), pars_cont, t_ode, "DDE"),
  run_model(pn_to_odin(sir, "stochastic"), pars_disc, t_disc, "Stochastic"),
  run_model(pn_to_odin(sir, "discrete"), pars_disc, t_disc, "Discrete")
)
#> i Using cached generator
#> i Using cached generator
#> i Using cached generator
#> i Using cached generator

ggplot(df_all, aes(time, I, colour = type)) +
  geom_line(linewidth = 0.8) +
  labs(title = "SIR model: four types from one Petri net",
       x = "Time", y = "Infectious (I)", colour = "Model type") +
  theme_minimal()
```




## Summary

| Type | `pn_to_odin()` | Use case |
|---|---|---|
| `"ode"` | `deriv()` | Standard deterministic dynamics |
| `"dde"` | `deriv()` + `delay()` | Fixed delays (incubation, immunity) |
| `"stochastic"` | `update()` + `Binomial()` | Demographic stochasticity |
| `"discrete"` | `update()` (Euler) | Discrete-time deterministic |

The Petri net is the single source of truth. Changing the model type is a one-line change to `pn_to_odin(sir, type = ...)`, and the stoichiometry remains consistent across all four formulations.